Free balancing equations worksheet worksheet 9

Maths | 7th grade | Algebra

1. If 5x - 3 = 2x + 9, what is the value of x?

A) 2
B) 3
C) 4
D) 5

2. The sum of three consecutive integers is 54. What is the smallest integer?

A) 16
B) 17
C) 18
D) 19

3. Which of the following is the solution to 3(y - 4) = 2y + 6?

A) 6
B) 8
C) 10
D) 12

4. If a rectangle's length is twice its width and the perimeter is 54 units, what is the width of the rectangle?

A) 6
B) 9
C) 12
D) 15

5. Simplify the expression: 7(2a - 3) - 5(a + 4).

A) 9a - 41
B) 9a - 29
C) 14a - 27
D) 14a - 35

6. If 4p - 7 = 5p + 3, find the value of p.

A) -10
B) 10
C) 6
D) -6

7. A number is multiplied by 3 and then 8 is subtracted to give 19. What is the number?

A) 7
B) 9
C) 11
D) 13

8. Solve for z: 2(z + 5) = 3z - 4.

A) 14

B) 8
C) 6
D) 2

9. If 9q + 4 = 7q - 6, what is q?

A) -5
B) 5
C) 10
D) -10

10. The area of a triangle is 24 square units. If the base is 8 units, what is the height?

A) 3
B) 4
C) 5
D) 6

Answer Key

1. C
Solve the equation: 5x - 2x = 9 + 3, which simplifies to 3x = 12, so x = 4.

2. B
Let the integers be x, x+1, and x+2. The equation is x + (x+1) + (x+2) = 54, simplifying to 3x + 3 = 54, or 3x = 51. Then x = 17.

3. A
Distribute the 3: 3y - 12 = 2y + 6. Simplify to y = 18. Correct answer is A.

4. B
Let width = w, then length = 2w. Perimeter = 2(w + 2w) = 54. Simplify to 6w = 54, so w = 9.

5. A
Distribute and combine like terms: 14a - 21 - 5a - 20 = 9a - 41.

6. B
Rearrange to 4p - 5p = 3 + 7, simplifying to -p = 10, so p = -10.

7. C
Set up the equation: 3x - 8 = 19. Solve to get 3x = 27, so x = 9.

8. C
Distribute and rearrange: 2z + 10 = 3z - 4, then z = 14.

9. A
Rearrange to 9q - 7q = -6 - 4, simplifying to 2q = -10, so q = -5.

10. D
Use the area formula: (1/2) * base * height = 24. So, 4 * height = 24, thus height = 6.